

Review

A comprehensive review of the applications of machine learning for HVAC

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ARTICLE INFO

Keywords:

Heating, Ventilation and air conditioning
Machine learning
Review
Optimization
Control
Fault detection and diagnosis

ABSTRACT

Heating, ventilation and air-conditioning (HVAC) accounts for around 40% of the total building energy consumption. It has therefore become a major target for reductions, in terms of both energy usage and CO₂ emissions. In the light of progress in building intelligence and energy technologies, traditional methods for HVAC optimization, control, and fault diagnosis will struggle to meet essential requirements such as energy efficiency, occupancy comfort and reliable fault detection. Machine learning and data science have great potential in this regard, particularly with developments in information technology and sensor equipment, providing access to large volumes of high-quality data. There remains, however, a number of challenges before machine learning can gain widespread adoption in industry. This review summarizes the recent literature on machine learning for HVAC system optimization, control and fault detection. Unlike other reviews, we provide a comprehensive coverage of the applications, including the factors considered. A brief overview of machine learning and its applications to HVAC is provided, after which we critically appraise the recent literature on control, optimization and fault diagnosis and detection. Finally, we provide a comprehensive discussion on the limitations of current research and suggest future research directions.

1. Introduction

Residential, commercial and other buildings account for around 40% of total global primary energy consumption and around 30% of the total related CO₂ emissions [1]. Reduced energy consumption in buildings is therefore a key component in achieving sustainable long-term development and lower carbon emissions [2,3]. The energy consumed by heating, ventilation and air-conditioning (HVAC) alone amounts to 35%–65% of the total energy, according to the type of building. A typical breakdown for public buildings is provided in Fig. 1. HVAC, therefore, provides a substantial source of energy-saving potential. With increasingly higher energy prices globally, there is an economic motivation behind reduced energy consumption, as well as the environmental benefits.

According to the International Energy Agency (IEA) [4], the peak power load of countries with low HVAC penetration rate is expected to increase by around 45% by 2050. This huge increase has aroused concerns over the management and energy efficiency of HVAC systems. At the same time, people may spend 60%–90% of their time indoors, and various factors such as thermal comfort, visual comfort, acoustic comfort and indoor air quality (IAQ) have a significant impact on the quality of life and comfort of people living indoors [5].

As economies develop, a greater emphasis is placed on indoor comfort, with higher requirements for HVAC control, while modern

buildings are becoming more dynamic and complex. From Fig. 2, it can be seen that an increasing number of households are installing air conditioning, which poses a challenge to the charge load, especially during peak periods of electricity demand, potentially compromising the safety of power grids. In addition to the main service systems of buildings, such as lighting and HVAC, more complex scenarios with on-site generation, energy storage systems and electric vehicle charging, are emerging. The stability and intermittency of on-site generation systems such as solar photovoltaics also presents challenges to the building service system. In order to achieve energy efficiency in the optimal operation of such grid interactive energy-saving buildings, flexible control methods are needed.

The optimization of HVAC mainly has two goals: energy saving and improving thermal comfort. Traditional energy saving methods can only improve the performance of equipment, and have little effect on energy savings, while involving huge investments. Thermal comfort involves many aspects, including air temperature, radiation temperature [7], airspeed [8,9], and relative humidity (RH), as well as personal factors, such as metabolic rate [10], clothing insulation [11], possible age [12], gender, and adaptation [13,14]. Traditional control methods struggle to achieve great improvements in thermal comfort. With the development of economies and societies, accompanied by greater ambitions and disposable incomes, societies demand greater levels of thermal comfort

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<https://doi.org/10.1016/j.decarb.2023.100023>

Received 1 July 2023; Received in revised form 20 September 2023; Accepted 20 September 2023

Available online 25 September 2023

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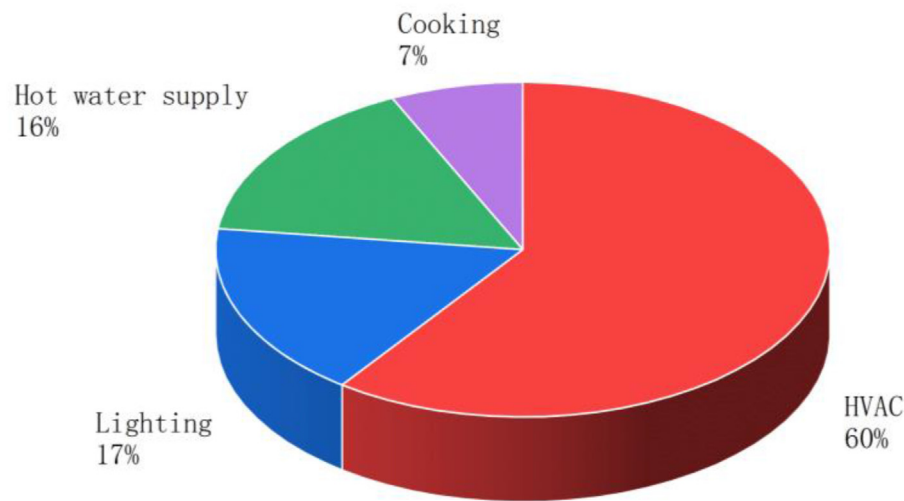


Fig. 1. Typical energy consumption composition of a public building [6].

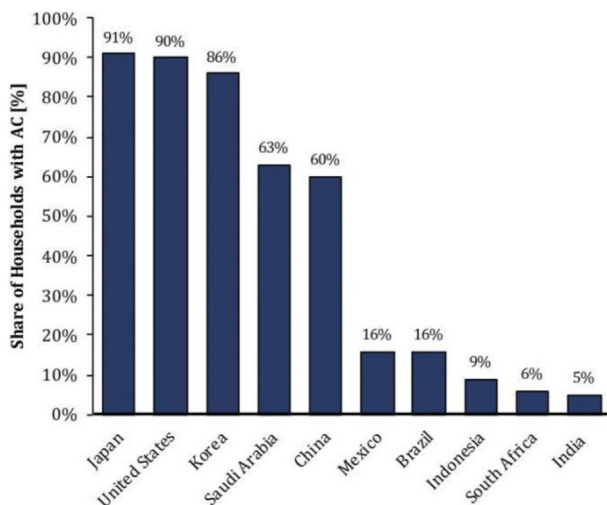


Fig. 2. Percentage of households equipped with air-conditioners in selected countries, 2018 [15]. Reprinted with permission from Elsevier.

while simultaneously developing a greater collective conscience with regards carbon emissions and energy conservation.

In order to resolve some of the issues associated with finding a balance between thermal comfort and energy savings, machine learning has received an increasing level of attention. Machine learning is a set of tools designed to extract information from data, reshape data, learn complex patterns, and make informed decisions. It forms the algorithmic core of artificial intelligence (AI). It can also learn from an environment and optimize performance from experience with a feedback loop involving rewards and penalties. Machine learning has made rapid progress in recent years, driven primarily by the greater availability of data, access to increasing computational power and budgets, as well as theoretical advances. There have also been developments in sensor technologies (enabling more accurate measurements), the Internet of Things (making data easier to collect and more widely available), cloud technology (making data storage more convenient and reliable), developments in hardware technology, such as CPUs and GPUs, and new or enhanced programming, especially GPU programming parallelization.

Alongside these developments there has been an enormous growth in the availability of open-source software, with the modularization of many machine-learning (including deep learning) techniques that can be called and combined within environments such as Tensorflow and PyTorch,

based on the Python language. Furthermore, there are a vast number of on-line and offline resources for learning how to implement machine learning, frequently free to access and providing codes or code snippets. The aforementioned data, high-performance hardware devices and advanced online resources have jointly promoted the development and application of machine learning methods in various fields, such as games [16,17], automatic driving [18,19], autonomous Internet of Things [20], intelligent buildings [21–24], smart cities [25], wireless networks [26], energy Internet [27], unmanned aerial vehicles [28], smart micro grids [29], edge computing [30], and manufacturing systems [31], to name but a few.

Machine learning methods were first applied to the control and optimization of HVAC systems around the 1990s. Since then, a great deal of research has been conducted on energy conservation and thermal comfort improvement. Many studies show that using machine learning methods to optimize HVAC can reduce energy consumption by 5%–30% and improve the comfort of residents. The most prominent machine learning methods in this area are artificial and deep neural networks (ANNs and DNNs), which are supervised. There are however, many other methods that can be employed, including support vector methods, Gaussian process models, kernel regression, dimension reduction techniques, and many others. Further discussion and classification of machine learning is provided in the next section.

The main applications of machine learning to HVAC can broadly be categorized as energy management and indoor environment control. The first of these determines an optimal energy management strategy by analyzing data from buildings together with outdoor meteorological data. It can be used to predict energy consumption and develop energy management policies with the goals of reducing energy consumption and financial costs. Optimal control of HVAC systems can also be achieved using machine learning, by analyzing indoor and outdoor conditions. This is a complex problem that may involve several competing objectives (energy consumption, costs, comfort), and the prediction of occupancy rates and behaviors.

Machine learning has also been used to a lesser extent for building design and planning, proposing better HVAC system solutions. A more active area is fault diagnosis, equipment monitoring and predictive maintenance of HVAC systems. Here, machine learning can be used to monitor and detect equipment faults and abnormalities, diagnose the root cause of problem and its severity, and for planning or scheduling maintenance.

There have been several literature reviews [15,32–43] on the combination of HVAC and machine learning in recent years, as summarized in Table 1. Some of these reviews only tangentially or indirectly cover machine learning or focus on specific machine learning techniques, and

Table 1
Summary of recent reviews on machine learning for HVAC systems.

Ref	Review content	Conclusion
Merabet Ghezlane Halhouli et al. [15]	A comprehensive and systematic review of AI-based techniques for building control, assessing their ability to increase the energy efficiency, while simultaneously maintaining thermal comfort.	AI techniques can open up opportunities for more energy savings while maintaining comfort. Adjusting dynamic setpoints in commercial buildings will depend on how comfort modeling is connected to human activity.
Hong Tianzhen et al. [32]	Systematic survey of how machine learning has been applied at different stages of a building life cycle.	Lack of large scale labeled data to train and validate models, lack of model transferability, lack of strong justification of costs and benefits of deploying machine learning, and the performance might not be reliable and robust for the stated goals.
Hasan Zahid et al. [33]	Comprehensive review and report on the modern applications of ML algorithms, including deep learning, transfer learning, active learning and reinforcement learning.	Deep learning algorithms are superior to traditional algorithms in decision making. Open challenges in terms of better performance and tackling unexplored issues in the cyber-physical building environment.
Esfafilian-Najafabadi Mohammad et al. [34]	Reviews research related to occupancy-based control systems. Classification based on integrating occupancy information with control systems. Identification of their strengths and limitations.	In many cases HVAC control systems based on user-defined occupancy schedules perform poorly with regards to energy savings and thermal comfort. Research gaps are identified and suggestions for objectives are put forward.
Moradzadeh Arash et al. [35]	A new hybrid machine learning model is proposed for forecasting the cooling load (CL) and heating load (HL) of residential buildings after analyzing various types of CL and HL forecasting models.	A hybrid model is superior for predicting building energy.
Maddalena Emilio T. et al. [36]	A review of the HVAC control problem for buildings is presented, emphasizing the distinguishing features.	Model predictive control especially combined with semi-parametric or non-parametric regression for identification is a promising alternative that needs further exploration.
Royapoor Mohammad et al. [37]	An historic and current perspective of building climate and plant control techniques. A survey is also reported.	More advanced computing technology (e.g., agent-based technology) has broad prospects.
Ghahramani Ali et al. [38]	Covers the necessity for integrated systems consisting of sensors, system interoperability infrastructure, learning/control, and actuators operating under a central intelligent system. Discusses current applications of intelligent personal thermal comfort systems in buildings.	More complex control algorithms should be implemented in the building management systems, to better manage the multitude of complex data inputs from various personal thermal comfort profiles.
Runge Jason et al. [39]	A review of the studies published since the year 2000 which have applied ANNs for forecasting building energy use and demand, with a particular focus on reviewing the applications, data, forecasting models, and performance metrics used in model evaluations.	The majority of the applications have been to commercial buildings, using hourly data, and applied to a whole building energy load. Errors of ANN models were found to be 0.001–36.5% (MAPE) for single step ahead forecasting and 1.04–42.31% (MAPE) for multistep ahead forecasting.

Table 1 (continued)

Ref	Review content	Conclusion
Yu Liang et al. [40]	A comprehensive review of deep reinforcement learning (DRL) for smart building energy management (BEM) from the perspective of system scales.	Almost all model-free DRL-based optimization methods have not implemented in practical BEM. Model-based DRL approaches are more practical. DRL-based energy management methods have the potential of improving some building performance metrics.
Mason Karl et al. [41]	A comprehensive review of the literature relating to the application of reinforcement learning (RL) to developing autonomous building energy management systems.	RL significantly improves the energy efficiency of homes. The vast majority of RL implementations for building energy management are in simulation-based. Much of the recent research focuses on deep versions of RL.
Han Mengjie et al. [42]	This paper conducts a comprehensive review of RL techniques applied in control systems for occupancy comfort in indoor built environments.	There is limited works on RL for controlling occupancy comfort, especially in indoor air quality and lighting. Relatively few studies have included occupancy patterns and/or feedback into the control loop. There is a gap with regards to the implementation of cooperative multiagent RL.
Yao Ye et al. [43]	Comprehensive review of applications of model predictive control (MPC) in building HVAC systems.	Proper implementation of MPC requires well-defined numerical values of control horizon, prediction horizon and sampling time. The modeling technique is considered one of the major crucial steps while implementing the MPC control scheme for building and HVAC control.
Chen Zhelun et al. [44]	This paper reviews and summarizes the literature on data-driven fault detection and diagnosis (FDD) from three aspects: process, systems studied (including the systems being investigated, the faults being identified, and the associated data sources), and evaluation metrics.	Yet significant challenges still remain for a wide market adoption of data-driven FDD methods. These challenges include real time and real-building implementation that is subject to data uncertainties; method performance benchmarking in real buildings; fault impact analysis; method scalability and transferability; fault interpretation; cyber security and data privacy; and user experience.
Zhao Yang et al. [45]	This paper aims at making a comprehensive literature review of artificial intelligence-based FDD methods for building energy systems in the past twenty years from 1998 to 2018, summarizing the strengths and shortcomings of the existing artificial intelligence-based methods, and revealing the most important research tasks in the future.	This paper analyzes the advantages and disadvantages of data-driven approach and knowledge-driven approach, and summarizes the important research tasks in the future. In the future, new approaches should be developed that combine the benefits of data-driven and knowledge-based approaches. An AI-based approach has great potential in solving FDD problems in building energy systems.

the majority are focused solely on control applications or more specifically on reinforcement learning for control. In this review we take a more holistic approach and review the broader applications of machine learning methods to the optimization, control and fault detection and

diagnosis for HVAC systems, as well as the different objectives considered, including comfort, energy consumption and costs. We provide an overview and brief introduction to the different types of machine learning, and the potential applications of machine learning in HVAC. We review the literature in detail and analyze the shortcomings of current research, putting forward suggestions for future research.

2. Methodology of review and search

We used multiple databases [46–48] to search for relevant literature with keywords “Machine Learning” * “HVAC system” * “in building”, identifying 221 papers. Preliminary relevance was determined according to the title. Further evaluation was based on the journal or conference, and abstract. The main principles for selecting papers were as follows:

- 1) Only papers published in peer-reviewed international journals or conferences are selected
- 2) HVAC systems located in buildings, excluding other environments, such as outdoor space and vehicles.
- 3) Only papers that use specifically machine learning methods to optimize the operation of HVAC systems.

Based on the above paper screening, we conducted a comprehensive review of core papers. For convenience, we provided a representative of some of the core papers in Table 2, along with key information. This information includes the type of building, machine learning methods used, whether to save energy or consider thermal comfort, the data sources, and comparison methods. Various types of buildings, especially offices, residential buildings, and campuses have been studied, with most studies considering energy conservation and a smaller number considering thermal comfort. In terms of energy conservation, a number of studies have provided energy-savings data, with claimed savings reaching 30% and over in some cases. Research data is generally measured (by sensors and other devices) or simulated from established models.

In the next section we first provide a brief overview of the potential applications of machine learning to HVAC, followed by a brief review of machine learning and the types of machine learning used for HVAC. We then review the literature in detail.

3. Machine learning and its applications to HVAC

3.1. Potential applications of machine learning for HVAC

Machine learning has great potential for improving the design and performance of building HVAC systems, aiding building managers in reducing energy consumption, improving indoor environmental quality, and reducing operating and maintenance costs. We now detail some specific applications of machine learning in this field.

a. Energy management

Machine learning can determine the optimal energy management strategy by analyzing the energy consumption data from buildings along with outdoor meteorological data. For example, it can be used to predict future energy consumption and develop more effective energy management policies to reduce energy consumption and costs using pure supervised machine learning or time series approaches based on machine learning.

b. Fault diagnosis, equipment monitoring and predictive maintenance

Machine learning can monitor the operating data of building equipment, detect equipment faults and abnormalities, and help to diagnose the root cause of problems. This allows for maintenance and repair to be carried out in a timely manner, ensuring the normal operation of equipment and extending its lifespan. For example, unsupervised

learning is able to detect equipment abnormalities, including a sudden temperature rise or drop, or equipment vibration.

c. Indoor environment control

Machine learning can be used for optimal control of HVAC systems by analyzing data interior and exterior to buildings, such as temperature, humidity, and CO₂ concentration. For example, machine learning can predict the number and behavior of indoor occupants, and adjust the operation of air-conditioning or heating systems accordingly to provide a more comfortable indoor environment. This, along with energy management are the areas in which the vast majority of applications of machine learning are to be found.

d. Building design and planning

Machine learning can be used for building design and planning, proposing better HVAC system solutions. For example, machine learning can help designers generate optimized design schemes by analyzing a large number of architectural design cases. It can identify which design elements or parameters have the greatest impact on the end result and guide the designer to make adjustments.

In summary, machine learning is widely used in building HVAC optimization, control and optimal control, at various stages from planning to operation and maintenance. This review mainly focuses on the application of machine learning for the reduction of energy consumption and improvement in occupancy comfort. In the next section, we will review the literature related to these two aspects in detail.

3.2. An overview of machine learning methods for HVAC optimization and control

The effectiveness of machine learning for HVAC and buildings has been widely demonstrated via a diverse range of applications. It is used for optimal control, fault detection and diagnosis during operation, building energy consumption prediction and energy savings optimization. Furthermore, it plays a huge role in the overall design of HVAC systems and intelligent buildings. There is a vast number and different categories of machine learning methods, and the choice of methods for a given task depends on the goal and the type of data available. It is also possible to combine different methods in a single study to achieve different goals or to perform sub-tasks towards a single objective. According to the nature of the data, we can divide machine learning into four categories: supervised learning, unsupervised learning, semi-supervised learning and reinforcement learning, as illustrated in Fig. 3.

The most common methods fall within supervised learning, which is based on data that takes the form of inputs and corresponding outputs (targets), arising from some process, system or even a model. The goal is to learn a mapping between the inputs and outputs, and this is achieved by devising a ‘statistical model’ of the process containing various parameters (or hyperparameters) that are estimated using (training) data. This estimation procedure, which we call ‘learning’, is based on either minimizing an error (loss function) between the predictions and targets, or maximizing a conditional probability of the targets, as a function of the parameters. In Bayesian learning, this approach is refined by seeking to sample (usually approximately) from a posterior distribution over the parameters, after appealing to Bayes’ law.

The outputs can lie on a continuous interval, leading to regression, or they can be categorical, leading to classification. In the latter case, the inputs are often said to be labelled, with the labels denoting different categories or classes, and from which corresponding numerical outputs can be defined. Common supervised learning methods includes linear regression, decision trees, support vector machine (SVMs), k-nearest neighbor (kNN) algorithms, artificial neural networks (ANNs), hidden Markov models (HMMs), naive Bayesian models, Gaussian mixture

Table 2
Summary of core applications of machine learning to HVAC.

Ref	Building type	Methods	Energy conservation savings	Thermal comfort considered	Data source	Factors	Comparisons	System
Wei T. et al. [21]	General construction	DRL, heuristic adaption	20–70%	Yes	Real solar, electricity price, climate data	Climate, electricity price, solar irradiation, occupancy	Baseline controller	HVAC
Zhao Huan et al. [49]	General construction	Hybrid-model-based RL	26.99%	Yes	Real data; simulation	Ambient temperature; electricity price; all zones temperature	Short term MPC/model-free	HVAC
Chen Bingqing et al. [50]	Multi-type building	MPC + RL	6.6%	Yes	Real data; simulation	Outdoor air temperature; Outdoor air Relative Humidity; solar irradiation; Occupancy, wind speed	RL	HVAC
Nagpal et al. [51]	Residence	MPC + Linear Matrix Inequality	No data	Yes	ASHRAE dataset, simulation	Ambient temperature; solar; indoor temperature	Nominal control	HVAC
Gao Guanyu et al. [52]	Laboratory	RL, DDPG, FNN	4.31%	Yes	SG-Singapore-Airp-486980; simulation	Climate, indoor temperature, solar, Wind speed	Base-line controller	HVAC
Yuan Xiaolei et al. [53]	Office	Rule Base Control + RL	7.7%/4.7%	Yes	Simulation, real climate data	Climate, indoor/outdoor temperature, electricity price	Rule Base Controller/PID controller	Variable Air Volume
Liu Boming et al. [54]	Residence	ANN, DDPG	No data	Yes	Simulation data, real climate, electricity price data	Climate, indoor/outdoor temperature, electricity price	Existing approaches	HVAC
Lee Seungjae et al. [55]	Office	Self-tuned (online learning + MPC)	No data	Yes	Measured data, real climate data	Air temperature, mean radiant temperature, air speed, humidity	Baseline MPC controller	HVAC
Mawson et al. [56]	Plant	ANN	No data	No	Measured data, simulation data	Production schedules, climate, temperature, humidity	None	HVAC
Deng Zhipeng et al. [57]	Office	ANN, RL, TL	Not considered	No	Measured and collected data	Indoor air temperature, relative humidity, thermostat set point, lighting occupancy, clothing level of occupants, occupancy	None	HVAC
Blad C. et al. [58]	Residence	RL + Q-Network and Q-Network + Eligibility Trace	No data	Yes	Measured data	Indoor temperature, ambient temperature	None	Underfloor heating system
Azuatalam Donald et al. [59]	Commercial building	RL + DR	22%	Yes	Real climate data	Climate, humidity, solar irradiation, occupancy	Default RL controller	HVAC
Yoon Young Ran et al. [60]	University	RL, Q-Learning, GP	No data	Yes	Collected data	Indoor/outdoor temperature, humidity, wind speed	Handcrafted baseline controller	HVAC
Aguilar Jose et al. [61]	Opera House	ML	No data	Yes	BMS data	–	None	HVAC
Hao Jun et al. [62]	University	Multi Agent RL, game algorithm	No data	No	eQUEST simulation data	Climate, occupancy	Original energy cost game algorithm	HVAC
Nasruddin et al. [63]	University	ANN, MOGA	No data	Yes	Measured data, real weather data	Temperature, humidity, building internal load, occupancy	Base-line controller	Two-chiller system
Wei Tianshu et al. [64]	Mixed-Use	Heuristic DRL based algorithm, ANN	26.9%	Yes	Real solar, electricity price data	Electricity price, temperature, equipment operation, occupancy, ambient temperature, solar irradiation	Base-line controller	HVAC
Yu Liang et al. [65]	Commercial building	Multi Agent DRL	No data	Yes	Measured data, Pacan street database	Occupancy, indoor/outdoor temperature, electricity price, CO ₂ concentration, climate	Baseline controller	HVAC
Ghahramani Ali et al. [66]	Office	KNN stochastic hill climbing, regression Decision Tree, RBS	31.17%	Yes	BMS database	Gas/electricity consumption, climate, occupancy	Baseline controller	HVAC
Pombeiro Henrique et al. [67]	Lab	DR + GA	No data	Yes	Weather station, Measured data	Dry-bulb temperature, solar irradiation, humidity, wind speed, electricity price	–	HVAC

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Table 2 (continued)

Ref	Building type	Methods	Energy conservation savings	Thermal comfort considered	Data source	Factors	Comparisons	System
Jang Ye-Eun et al. [68]	Office	ANN, online learning	No data	Yes	BEMS database	Electricity price, climate, temperature,	Traditional control method	HVAC
Brandi Silvio et al. [69]	Office	DRL	5–12%	Yes	Measured data, real climate data	Climate, occupancy, different indoor temperature setpoint	Baseline controller	heating system
Lork Clement et al. [70]	University	DRL	No data	Yes	Real climate data, collected data	Indoor/outdoor temperature, climate, AC setpoint/power status, humidity	Rule-based controller	AC
Valladares William ey al [71].	Lab, classroom	DRL	4.5%	Yes	Real climate data, simulation data	Indoor temperature, levels of CO ₂ , humidity, occupancy	Current control system	ACV
Du Yan et al. [72]	Residence	Multi-task DRL	No data	Yes	Real climate/ electricity price data	Indoor/outdoor temperature, electricity price, climate	Rule-based controller, single task method	HVAC
Gupta Anchal et al. [73]	Residence	DRL	5–12%	Yes	Real climate data	Indoor/outdoor temperature, climate	Traditional thermostat controller	Heating System
Li Xuan et al. [74]	Office	CNN	No data	Yes	Tested and collected data	Indoor temperature, clothing level, thermal sensation	None	HVAC
Jiang Zhanhong et al. [75]	Office	DRL	No data	Yes	Measured data, typical Meteorological Year data	Electricity price, indoor/outdoor temperature, climate, cool load	Two simple rule-based policies	HVAC
Du Yan et al. [76]	Residence	DRL + DDPG	15%	Yes	Real weather data	Indoor/outdoor temperature, electricity price	DQN, rule-based controller	HVAC
Esrafilian-Najafabadi, M. et al. [77]	Residence	MLP	No data	Yes	Gathering from five residential units	Occupancy, indoor/outdoor temperature, climate, solar irradiation, wind speed	Rule-based controller	HVAC
Kim Y. J [78].	Multi zone office	DR + ANN	No data	Yes	BEMS database	wind speed, solar insulation, thermal load, occupancy, electricity price, ambient temperature	Ideal physics base, non-DR strategy	HVAC
Yoon A. Y. et al. [79]	Commercial building	ANN, DR	No data	Yes	Collected data	–	None	HVAC
Chen Yujiao et al. [80]	Residence	RL	13–23%	Yes	Real climate data	Indoor/outdoor temperature, humidity, solar irradiation, wind speed, windows operate, occupancy	Rule-based heuristic control strategy	HVAC
Chen Yujiao et al. [81]	Smart buildings	DNN, TL	Not considered	No	Simulation and collected data	–	None	HVAC, natural ventilation
Zhang Zhiang et al. [82]	Office	BEM-DRL	No data	Yes	Real world data	Indoor/outdoor temperature, humidity, solar irradiation, wind speed	Rule-based controller	Radiant heating system
Park Jaesung et al. [83]	Residence	ML	8–10%	Yes	Simulation data	Indoor/outdoor temperature, humidity, solar irradiation, climate	Traditional PMV controller	HVAC
Mawson Victoria Jayne et al. [84]	Manufacture	DQN	No data	No	Simulation data	Climate, occupancy, manufacture schedules, humidity, wind speed	None	HVAC
Hao Cheng Zhu et al. [85]	Office	ANN	35%	Yes	CFD Simulation data	Indoor CO ₂ concentration, temperature, humidity	Traditional controller	HVAC
Radhakrishnan Nikitha et al. [86]	Office	Learning Base Token Scheduling Algorithm + MPC, adaptive learning	5.7%	Yes	Simulation and sensor measurement data	Indoor/outdoor temperature, occupancy	Centralized optimization and Token Base Scheduling Algorithm approach	HVAC

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Table 2 (continued)

Ref	Building type	Methods	Energy conservation savings	Thermal comfort considered	Data source	Factors	Comparisons	System
Asad Hussain Syed et al. [87]	Supertall	Adaptive Model Base Real Time Optimization + Hybrid Genetic Algorithms	8%	Yes	Real-time measurements	Temperature, humidity, wind speed	Conventional fixed set point and MRTD approaches	HVAC
Wang Junqi et al. [88]	Skyscraper	Event Driven Optimal Control	0.9–4.6%	Yes	BMS, field measured data	Climate, temperature, humidity, wind speed	Traditional time-driven optimal control	HVAC
Lymperopoulos G. et al. [89]	Primary school	Adaptive control	No data	Yes	Measured data, official data	Climate, human activity, lighting, electrical equipment, door open and closed	Distributed constant-gain control	HVAC
Vosughi A. et al. [90]	Residence	MJLS	Not considered	Yes	Internet Of Things data	Solar irradiation, climate, indoor heat gain, human activity	Control base on thermostat in fixed location	HVAC
Dullinger Christian et al. [91]	Multi-type building	MILP-MPC	No data	Yes	Real world dataset	–	None	HVAC
Kumar Ranjeet et al. [92]	University	Stochastic MPC	7.5%	Yes	No	Electricity load, cooling and heating load, electricity price	Stochastic MPC	HVAC
Anuntasethakul C. et al. [93]	Office	MPC + SVM	12.46%	Yes	Measured and simulation data	Temperature setpoint, wind speed, outdoor temperature, heating load	Traditional control	HVAC
Ellis Matthew J. et al. [94]	Residence	An encoder–decoder LSTM-based EMPC framework	No data	Yes	Measured data	Indoor/outdoor temperature, temperature setpoint, solar irradiation, internal load	None	HVAC
Ahn Ki Uhn et al. [95]	Office	DQN	15.7%	Yes	Measured and simulation data	Occupancy, lighting, equipment operation, CO ₂ concentration, temperature	Baseline controller	HVAC
Wang Shiqiang et al. [96]	Water cube	Decentralized Swarm Intelligence Algorithm	6%	Yes	Measured data	–	Traditional method	HVAC
Gao Guanyu et al. [97]	Residence	DRL	No data	Yes	Sensor measurement data, real climate data	Indoor temperature, humidity, mean irradiation temperature, air speed, clothing, insulation	Baseline controller	HVAC
Fu Chenhui et al. [98]	Multi-zone	TD3-MPC	16%	Yes	Simulation data	Electricity price, indoor/outdoor temperature, outdoor flow rate	Other DRL methods	HVAC
Martinez-Garcia Fernando et al. [99]	Data center	WRLS, General Predictive Controller	15%	No	Simulation data, measured data	–	Nominal Adaptive Predictive Controller	Cooling unit
Mirhoseininejad S. et al. [100]	Data Centers	Data-driven, MPC	No data	No	Simulation data, measured data	–	None	Thermal Control System
Xie Di et al. [101]	University	MPC + Alternating Direction Method Of Multipliers	No data	Yes	Real world data	Electricity price, outdoor/indoor temperature, most comfortable temperature	Traditional MPC	HVAC
Li Zhiwei et al. [102]	Multi-zone buildings	Improved Particle Swarm Optimization, Newton-Raphson + MPC	No data	Yes	Simulation data	–	None	HVAC
Chi Zhang et al. [103]	Data center	Model-base DRL	17.1–21.8%	Yes	Real world data, simulation data	Outdoor/indoor temperature, equipment electric, climate, demand power	Baseline controller	HVAC
Chi Ronghu et al. [104]	Multi-zone	Distributed Iterative Learning Temperature Control	No data	Yes	I/O dataset	Climate, outdoor/indoor temperature, occupancy	None	HVAC

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Table 2 (continued)

Ref	Building type	Methods	Energy conservation savings	Thermal comfort considered	Data source	Factors	Comparisons	System
Shen Gulai et al. [105]	Apartment	MPC + data-drive	33.2%	Yes	Real world EWH dataset	Climate, temperature, electricity price	Traditional control	Water heater system
Costanzo G. T. et al. [106]	Lab	Bench RL + Fitted Q-iteration	No data	Yes	Simulation data	–	None	HVAC
Yang Inho et al. [107]	Office	ANN	No data	Yes	Simulation data	Indoor temperature, air variation rate	Traditional control	HVAC
Zou Zhengbo et al. [108]	Office	DRL + LSTM	27–30%	Yes	BAS database	Indoor temperature, climate, CO ₂ concentration, humidity, wind speed	Rule-based controller	AHU
Ikeda S. et al. [109]	Hotel	Meta heuristic, ANN	13.4%	Yes	Society of Heating, Air-Conditioning and Sanitary Engineers of Japan	Outdoor/indoor temperature, solar irradiation, electricity price	Traditional control	HVAC
Amin U. et al. [110]	Office	Price-based DR + ANN	No data	Yes	–	Outdoor/indoor temperature, solar irradiation, electricity price, occupancy	Traditional control	HVAC
Afroz Zakia et al. [111]	Residence	Nonlinear Autoregressive Exogenous + PSO	No data	Yes	Measured data, real world data	CO ₂ concentration, indoor air temperature setpoint, solar irradiation, outside wind direction	Traditional control	HVAC
Wang Zu et al. [112]	Residence	ANN	No data	Yes	Measured data	–	Traditional control	HVAC
Deng Xiangtian et al. [113]	Office	Non-stationary DQN	No data	Yes	Typical meteorological year data	–	State-of-art DQN method	Variable Air Volume
Rana Mashud et al. [114]	Commercial buildings	Regression forest	Not considered	No	Real world data	Cooling load, ambient temperature	Gradient Boosting Regression	HVAC
Qin Hao et al. [115]	Office	SVR, RL	No data	Yes	Measured data, real world data, simulation data	Temperature, humidity, level of CO ₂ , number and location of people inside the room	None	HVAC
Cho Jihyeon et al. [116]	School buildings	Integrated neural network	No data	Yes	Simulation data, measured data	Indoor CO ₂ concentration, indoor PM10 and PM2.5 concentrations, temperature, humidity	Traditional control	HVAC
Zhuang Dian et al. [117]	Office	RL, time-series forecasting	No data	Yes	Measured data	Air temperature, relative humidity	None	HVAC
Faiz Muhammad Faizan et al. [118]	Residential building	Transformer Neural Networks, CNN-LSTM	No data	Yes	Measured data	Anthropometric measurements, air quality, power quality, indoor environment ambiance, and meteorological data	None	HVAC
Cho Ji Hyeon et al. [119]	School building	ANN, MOGA	Not considered	No	Simulation data	Thermal environment, life-cycle cost of building, energy	Optimal model based on general multi-input-multi-output	HVAC
Wang Bo et al. [120]	Office	Deep learning with Meta-heuristic based algorithm	No data	No	Simulation data	HVAC operating data	Other ML methods	HVAC
Ra Seon Jung et al. [121]	Factory building	Measured data	No data	Yes	Measured data	Air temperature, relative humidity, life-cycle cost	Traditional control	HVAC
Fu Qiming et al. [122]	Office	Multi-Agent	No data	Yes	Simulation data, history data	Climate, real-time cooling load and outdoor wet-bulb temperature	Rule-based control method	Cooling Water System Control
Zhong Xuyang et al. [123]	Office	End-to-End DRL	No data	Yes	Measured data	Weather and indoor environment condition	Baseline control method	HVAC

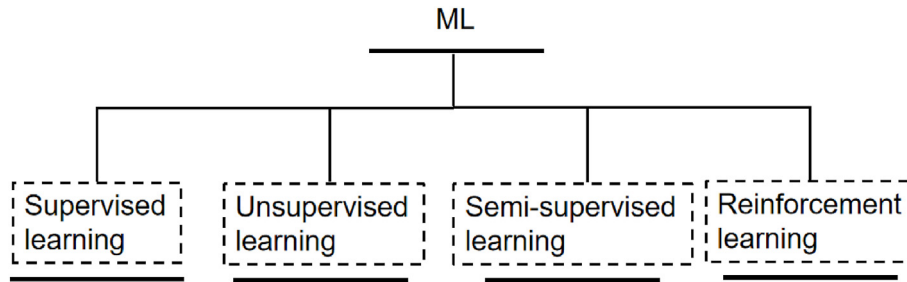


Fig. 3. Classification of machine learning methods.

models (GMMs), and Gaussian process (GP) regression (GPR) and other GP models in which the data is assumed to arise from a GP or mixture of GPs; these are only a small fraction of the methods available.

Compared with supervised learning, unsupervised learning does not involve any outputs or labelled inputs; the data comprises a set of singular data points. The process of learning is similar to that for supervised methods, but the goal is now to infer or extract hidden internal structures within the data that can reveal important information. As well as extracting important features underlying the data (feature extraction or representation learning), which are essentially some linear or nonlinear combinations of the data coordinates (attributes), unsupervised learning can also group data points according to different criteria (clustering), or reduce the size of the data set, i.e., find an approximating subspace (dimension reduction). Unsupervised learning is often used for data preprocessing before the application of supervised learning. Semi-supervised learning can be considered as a fusion of supervised and unsupervised learning. It involves a relatively small portion of labelled data and a relatively large volume of unlabeled data. In principle the latter can be labelled, but it is frequently too time-consuming to do so. The labelled data can be employed to train an initial model, which is subsequently employed to make predictions for the unlabeled data. This can be achieved in different ways, e.g., confident predictions on unlabeled data are added to the labeled data to repeat the training process (self-learning).

Compared with the static learning of supervised and unsupervised methods, reinforcement learning constructs artificial agents that take different actions in a given environment in order to learn dynamically, by maximizing cumulative rewards. Its biggest feature is learning interactively. Reinforcement learning is mainly composed of the agent, its environment, state, actions and a reward mechanism, as illustrated in Fig. 4. After the agent executes an action, the environment attains a new state, and a reward signal (positive or negative) is assigned to the action taken. Subsequently, the agent executes a new action according to the current state and the reward feedback, with the ultimate goal being to learn a policy by maximization of a cumulative reward, in which the

policy refers to the map between the states of the system and the action taken, and is usually assumed to be a probability distribution characterized by certain parameters. The value function represents a value for a certain state, or the expected return from a given state. Usually, the evolution is defined in terms of a Markov decision process.

In the RL framework there are transition and reward functions, together which form the dynamic model. In so-called model-based RL the dynamic model is learned first, usually offline. Conceptually, therefore it is similar to the system identification step of model predictive control, described below. The development of a system model is highly time-consuming, and this must be weighed against the potential benefits in terms of accuracy. In model-free RL, the transition and reward functions are not learned a-priori. Here, there are several choices, the main being value-based, actor-critic and policy gradient approaches. In policy gradient, the idea to update the policy parameters by using the gradient of the expected cumulated reward, with respect to the parameters. In value-based RL, a parameterized value function is instead estimated, from which the policy can be found. The actor-critic approach works with both the policy and the value function. As in policy gradient, the policy parameters are adjusted according to a gradient, but this time the gradient of the value function with respect to its parameters, rather than that of the cumulative reward.

Deep reinforcement learning (DRL) uses DNNs for policy learning and has gained popularity in recent years. A number of DRL algorithms are available, including Deep Deterministic Policy Gradient (DDPG), Deep Q-Networks (DQN), Proximal Policy Optimization (PPO) and Trust Region Policy Optimization (TRPO).

There are, moreover, other types of learning, which are used with one of the aforementioned four categories. Two prominent examples are transfer learning (TL) and multi-task learning (MTL). In TL, data from a similar or identical task to that of the target task is used to pretrain a model (find initial values for the parameters and therefore specify the latent mapping) before training on the task-specific data. The inputs may be the same for the two tasks, while the target map is different, or the inputs are different, while the target function is the same. It may also be that both the inputs and target map are different. This type of learning can help to vastly improve accuracy and is widely employed. MTL refers to the learning of multiple tasks simultaneously, as opposed to independently. These tasks, along with the associated data originate from the same process, and are therefore correlated. MTL attempts to incorporate these correlations in the joint learning, which can help to improve the overall accuracy.

In HVAC optimization and control, the machine learning approach generally requires prior knowledge or labelled data. The vast majority of algorithms are therefore either supervised learning or reinforcement learning, with semi-supervised and unsupervised learning rarely employed. There are some specific areas in which machine learning has been deployed in HVAC, which we now briefly describe before moving onto a detailed discussion of the literature in the next section.

Advanced control algorithms for buildings can effectively achieve energy savings and reduce carbon emissions. For HVAC, model-based predictive control (MPC) has therefore gained much popularity, although in reality the up-front time and cost of these approaches has

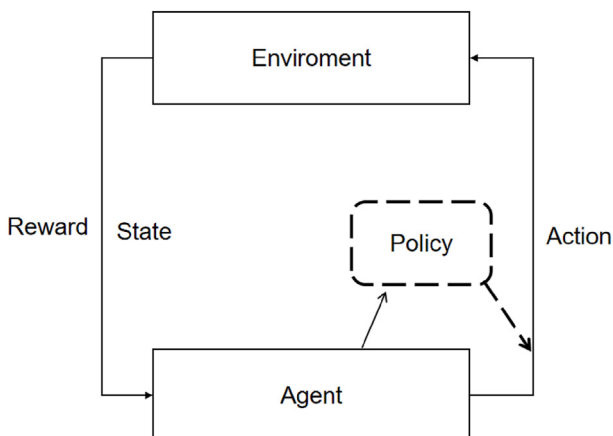


Fig. 4. Reinforcement learning process.

prohibited their practical deployment, with traditional rule-based methods continuing to be favored. MPC can in theory optimize HVAC systems, accounting for operating conditions, weather forecasts, electricity prices, and other factors, while simultaneously ensuring occupancy comfort and any operational and safety constraints. MPC relies on the use of an accurate HVAC model, and there has been a growing trend to replace physics-based with black-box (ML based) models due to a number of inherent difficulties with the former. The optimization objectives of HVAC systems are to reduce energy consumption and maintain comfort, which are conflicting. An MPC strategy can ensure an optimal balance between these conflicting objectives.

MPC mainly comprises three components (illustrated in Fig. 5): modeling, prediction and control. In HVAC optimization, modeling refers to the dynamic building model. Prediction refers to predicting the system state in the future based on the current state and control inputs. Control refers to maintaining the predicted system state as close to the expected value as possible.

The performance of traditional MPC control methods is highly dependent on the accuracy of the physical model. In order to improve accuracy, the usual approach is to increase the complexity of physical models or to use more sophisticated nonlinear optimization solvers. However, more accurate physical models rely on detailed, quantitative knowledge of the physical process and associated system parameters, which is not always available, especially in uncertain environments. Such models, when they are physics based, contain several thousands of parameters. Moreover, complex physical models and nonlinear optimization lead to higher online computational complexity and storage requirements, limiting their applications outside academic research. With different buildings having different characteristics, system and parameter identification is required for each new building, while the system parameters change over time, demanding some form of adaptive control that updates parameters online.

Training offline ML or algorithms (as building models) is much faster than physics-based modelling. It is also possible, in principle, to rapidly update models online as new data becomes available. ML, however, also possesses drawbacks. Its predictive performance is highly dependent on the number of training points, and in many application fields such data can be difficult to obtain. Moreover, most black-box methods (especially deep learning) are not explainable, meaning that observed trends in predictions cannot be easily identified with physical laws and processes. ML in general does not provide performance guarantees, with the exception of probabilistic approaches such as GP models. This is a major drawback, especially in safety-critical applications. Accordingly, it is not difficult to see that a combination of machine learning and MPC that

mitigates the shortcoming and leverages the advantages of each has broad prospects for optimal control of HVAC. Reinforcement learning has also gained much attention for optimal control, as an alternative to MPC, especially DDPG, which uses an offline training procedure to establish the policy.

3.3. HVAC control and optimization using machine learning

In this section, we will discuss in detail the application and performance of machine learning methods in HVAC optimization and control. The optimization of HVAC systems is mainly aimed at reducing the energy consumption of HVAC and improving the comfort of users. Some studies consider peak electricity prices to reduce the cost to users. A great deal of research has been carried out on machine learning within this context. A breakdown of machine learning and other methods for optimization and control used in the research literature is provided in Fig. 6. Some papers based on hybrid methods use more than one of these techniques.

As explained in Section 3.2, it is very difficult to establish a reliable and transferrable thermodynamic model in HVAC. In Ref. [53], the authors therefore use a reinforcement learning algorithm, combined with a rule-based control method, so that engineering expertise can be used to define conditional judgment statements. This allows for the determination of reasonable ranges of system control variables under various working conditions. At the same time, a RL algorithm collects and analyzes the operational data (i.e., interacts with its environment), continuously improving the control strategy and optimization of the system. RL has been shown to be a possible approach for HVAC systems [124] but its advantages are offset by the requirement of increased sample complexity, requiring enormous amounts of data (interactions with the environment) to compete with a simple feedback controller [125], even with pre-training [126]. The long training time may explain the limited number of real-world applications of RL for HVAC control. In order to make feasible the approach, high-fidelity system models are required for pretraining of the agents [125,127], which introduces challenges in terms of simulation-to-real data transfer and once again places the emphasis on realistic models of the process.

To solve the problem of sample complexity, Li et al. [128] as well as Jia et al. [126] used historical data from an existing controller to initialize the agent, as well as using the existing policy as a baseline [128]. In contrast, others [129,130] employed a model of the system dynamics in the RL framework, while Costanzo et al. [106] expanded the replay memory data using state-action pairs generated from a neural network, with batch-processing RL. In the approach of Nagy et al. [130] a neural network was trained to act as a model for state transitions, and multi-step planning was built into the RL. Model-based RL was found to be better overall than a model-free RL when taking into account sample efficiency and the end results (efficiency and comfort). A similar approach was adopted by Chen et al. [129] using an input-convex network developed in Ref. [131], validated using data from large commercial office buildings (16 thermal zones with temperature setpoints), although they only considered efficiency and not thermal comfort.

To overcome the limitations of model-dependent RL and sample efficiency, Chen et al. [50] combined RL with a differentiable MPC strategy [132]. No prior knowledge is required in their approach, and only historical data is used to train the model. The method was tested on 3 different scenarios and was shown to be better than a generic neural network strategy. Similar to Refs. [126,128], the authors were able to avoid the need for a high-fidelity model by initialization (based on data from an existing controller).

Azuatalam et al. [59] considered both energy savings and thermal comfort while implementing demand response, which presents challenges in terms of calibrating a systems model to capture the demand response. The authors employed a whole-building model and did not apply the algorithm to a real building. In Ref. [106], Shen et al. also incorporate demand response, for electric water heating (EWH)

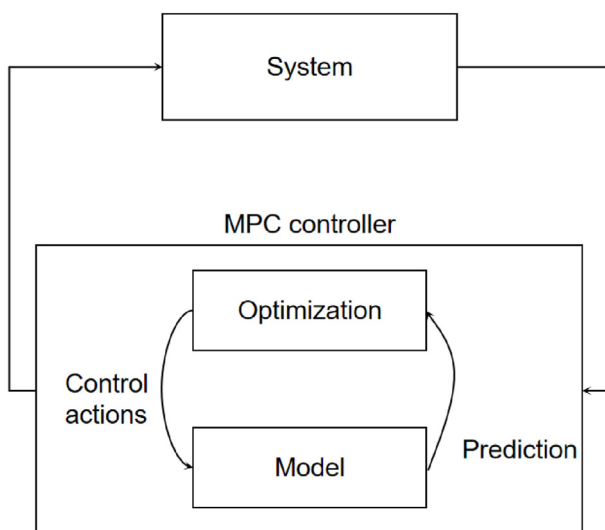


Fig. 5. MPC strategy and flow of information.

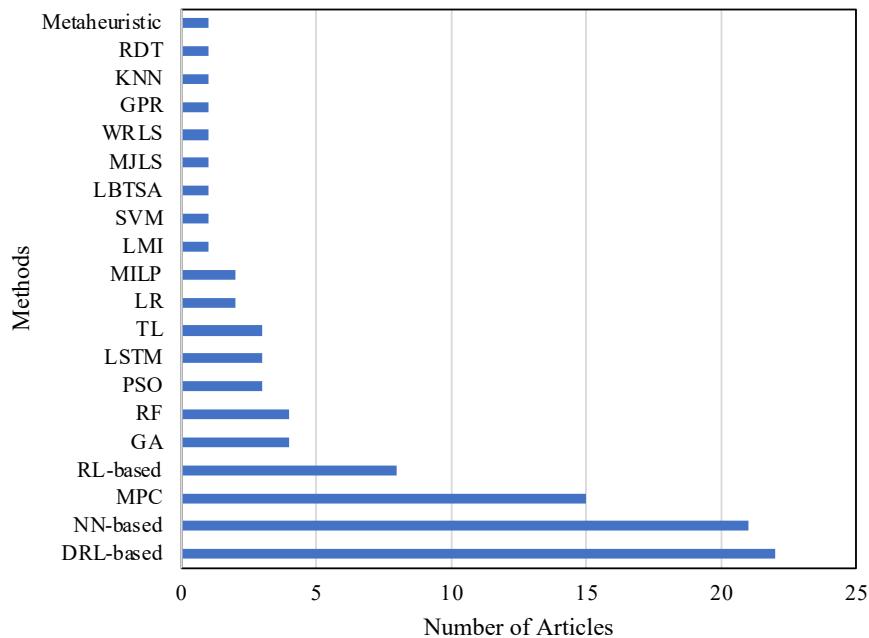


Fig. 6. A breakdown of machine learning and allied algorithms used in the literature.

scheduling and control using a combination of data-driven forecasts in a robust MPC with an EWH model, applied to a domestic water heating system in a multi-unit apartment building. The uncertainty interval of the model was able to anticipate demand to within 97% of the actual demand and electricity costs were reduced by up to 33.2% without affecting user comfort. The obvious drawback is again a reliance on a non-transferrable model.

Hao et al. [62] developed a novel multi-agent RL algorithm for optimal control of HVAC and planning of a power system, with each agent assigned to control a central HVAC system in each building. In this way it is possible to analyze the relationship the different HVAC control systems, and for the overall system to utilize data from a stochastic power grid, as well as weather conditions and human activity. Moreover, it can exploit past experiences to propose optimal HVAC settings. In Ref. [70], Lork et al. used aggregated data from households to train a neural network before devising individual models for the households in order to reduce data imbalances (some having more than others) as well as ameliorate overfitting. Specifically, they used a Bayesian CNN to model air conditioning in order to account for uncertainty in the planning. They then used a Q-learning reinforcement for AC control incorporating uncertainty arising from the models.

Chen et al. [80], used a Q-learning RL strategy to control natural ventilation with a view towards both efficiency and comfort, and found it to be superior to a heuristic control strategy, although there was no comparison to MPC. Q-learning is again used to train a RL algorithm in Ref. [58], with the incorporation of eligibility trace to account for the slow thermodynamics of an underfloor heating system. There are no comparisons with other methods so the performance is difficult to judge. Although reinforcement learning has certain advantages in HVAC control, its disadvantages are also obvious. First, reinforcement learning needs a rich database. Second, it is not efficient and cannot be directly used in other systems. In Ref. [57], Deng and Chen used Q-learning with occupant behavior modelled as a Markov decision process, alongside transfer learning (TL) in an attempt to apply it to other buildings, with moderate success. It is worth pointing out that in HVAC applications, RL is almost always in the form of Q-learning.

Biemann et al. [133] compared the performance of several RL methods, including a soft actor critic, twin delayed deep deterministic

policy gradients (TD3), trust region policy optimization (TRPO) and near end policy optimization (PPO), in terms of energy conservation, thermal stability, robustness and data efficiency. The soft actor critic algorithm achieved stable performance with 10 times less data than other strategy methods. This study suggests that this algorithm should be explored in future experiments, especially real-world experiments, with more comparisons.

ANNs have been used in a number of studies [54,83,85] to predict indoor conditions in order to control indoor environments and energy consumption. For example, in Ref. [100], the authors use an ANN for temperature predictions in a thermal model inside a MPC strategy for power efficiency. Similarly, connected ANNs in the form of LSTMs were used for each subsystem in a temperature set-point scheduling strategy developed by Kim [68]. In this method, the influence of electricity price is considered. There are relatively few studies on the key influencing factor of electricity price for reducing cost. In general, there are many cases where DR is used to consider electricity price.

Yoon et al. [79] combined ANN and DR strategies to consider the optimization of electricity price. An optimization problem of price-based DR was established using an ANN, and was solved repeatedly to generate optimal HVAC load plan data for different electricity prices and building environments. Subsequently, another ANN is trained online to directly predict the optimal load dispatch. In this study, a single-layer decision-making structure is used to achieve the optimal retail price, which makes the pricing optimization problem simple and increases the scalability of the model. Amin et al. [110], used an ANN to establish a thermal response model of the building and integrate it into an optimal HVAC scheduling. In addition, the different thermal preferences of residents were expressed as a coefficient of the bid price (selected by the occupants) in response to the price signal.

In solving nonlinear and multi-objective HVAC optimization problems, GA algorithms are extremely popular. It is a global optimization algorithm that simulates natural evolution, and crucially is gradient-free so that it does not require expensive numerical differentiation. On the other hand, its use is primarily restricted to complex multi-objective and high-dimensional optimization problems since it often sacrifices accuracy for the sake of finding a global optimum. GAs have been successfully applied to HVAC system control [134], green building design [135] and

thermal performance of a refrigeration system [136]. In Ref. [63], Nasraddin et al. combine an ANN with a multi-objective GA algorithm to optimize a double cooling unit system with the percentage of people dissatisfied (PPD) and annual building energy consumption as the objectives. In Pombiero et al. [67], the authors compare dynamic programming with a thermal model to a GA with the same thermal model and finally a GA with EnergyPlus. The objective is to maximize user comfort while minimizing costs, with constraints related to thermal comfort and available electricity (from a PV/battery system), and cost. The GA with EnergyPlus yielded the best performance, but interestingly dynamic programming with a thermal model outperformed the GA with the thermal model. This is certainly worthy of further investigation, although the authors say little about this conclusion.

Neural networks have a number of shortcomings, such as a large number of parameters, high model variance, a danger of overfitting, and a lack of systematic methods for architecture and hyperparameter selection, all of which compromise generalization. Although DNNs [137], which are essentially ANNs with more than one hidden layer, share these drawbacks, they are unrivalled in pattern recognition problems. The success depends on access to large volumes of data so that accurate learning is possible. Increasingly, DNNs have become more popular for HVAC applications compared to traditional ANNs [138–141]. A deep MLP was used by Ikeda et al. [109] to predict the optimal operation of an integrated cooling tower system, and a meta heuristic method was used to optimize the operation of other components. The use of a DNN reduced the daily operational costs compared to a conventional scheduling strategy, although few comparisons were made.

Particle Swarm Optimization (PSO) has attracted attention for its advantages of easy realization, high precision and fast convergence. Like GAs, PSO algorithms are population-based methods. Multi-objective particle swarm optimization algorithms have been used in several studies [61,96,111] with good results. In Ref. [77], using historical measurements of outdoor weather, indoor temperature and HVAC system lag time, a deep MLP was developed to estimate the dynamic pre adjustment time of residential buildings. A novel control strategy is proposed by Li and Chen [74] to adjust the set value of the thermostat according to the clothing level and the average facial skin temperature. In this study, a new CNN model is established for clothing classification by using TL, and the clothing level of residents is measured in real time. TL plays an important role in machine learning in HVAC optimization and control. In practice, the target building may not have enough available data, and the combination of TL and DNN can help to essentially augment the data set. Potentially, TL can be used to provide predictions for a new target building accurately using limited operational data, which can greatly reduce the workload.

Supervised learning can have a strong function approximation capability, but lacks decision-making ability. Reinforcement learning, on the other hand, has good decision-making ability but is unable to deal with perceptual problems. Therefore, a combination of the two provides a solution to the perception-decision-making problems relating to complex systems. As illustrated in Fig. 7, deep reinforcement learning (DRL) is an end-to-end perception and control system with strong universality. The learning process can be described as: at each moment, the agent interacts with the environment to get a high-dimensional observation, and uses a DNN to perceive the observation, so as to obtain the specific state feature representation; evaluate the value function of each action based on the expected return, and map the current state to the corresponding action through some strategy; the environment reacts to this action and obtains the next observation. Through repetition of the above process, the optimal strategy to achieve the goal can be accomplished.

In [21], Wei et al. applied DRL to HVAC control for the first time in the form of deep Q-learning (DQN). By constructing a DNN to associate the estimated value with the associated state action pair, a large state space can be processed, thus overcoming the drawbacks of traditional state action association algorithms. In Ref. [73], Gupta et al. attempted to enhance heating control through DRL, but it used simple

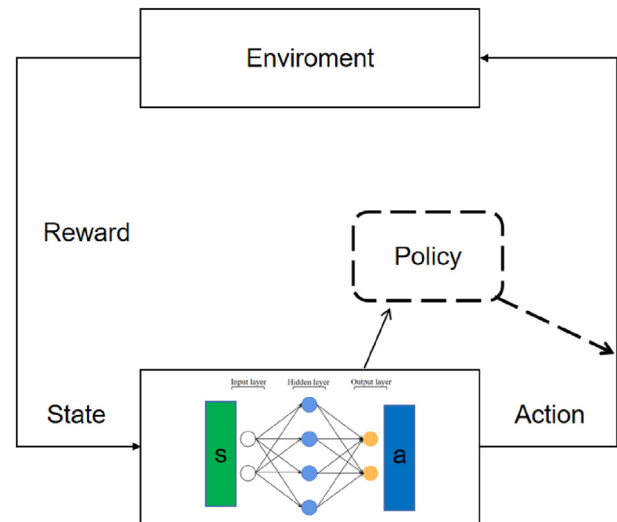


Fig. 7. Deep reinforcement learning optimization process.

first-order linear simulation experiment, and the simulation results were not convincing. Valladares et al. [71], used an improved DRL strategy based on DQN to optimize the consumption of energy, taking into account thermal comfort, energy consumption and indoor air quality. A HVAC control algorithm based on a multi-agent DRL was adopted in Ref. [65] for minimizing costs in a multi-zone commercial building subject to dynamic prices, taking into account occupancy, thermal comfort, as well as air quality comfort. In Ref. [69], Brandi et al. highlight the importance of selecting the hyperparameters in DRL algorithms, though this is hardly an issue confined to HVAC. As stated earlier, tuning the hyperparameters and equally as important the architecture involve an enormous number of choices for complex networks. State-of-the-art randomized grid searches are far from able to search the entire design space, but at the present time are the best available options.

Du et al. [72] proposed a MTL DRL in the form of deep policy gradient (DDPG) for both cooling and heating tasks simultaneously, in contrast to most approaches that focus on either one or the other. Comparisons to a single-task DDGP showed the MTL approach to be superior in terms of cost savings and user comfort. In Jiang et al. [75], under the premise of minimizing the energy cost under time-varying electricity price and demand electricity price, DQN with an action processor was developed, and a reward shaping mechanism was developed to overcome the problem of reward sparsity caused by demand charging. In Ref. [64], Wei et al. developed a DRL based heuristic algorithm to effectively distribute interactive workloads among geographically distributed mixed-use buildings (MUBs), thereby improving their energy efficiency.

Zou et al. [108] proposed a framework to achieve optimal control of three air handling units (AHUs), utilizing an LSTM network to approximate the real HVAC operation, and combining it with a DDPG approach to achieve optimal control of the AHUs. In Ref. [95] Ahn et al. adopted DQN to improve the control strategy of a HVAC system based on prior behavior, status and reward. This study proposes a combination of TL with DRL for generalization to other buildings and enhance the self-learning ability, as did Deng and Chen [57]. Yoon and Moon [60] studied an integrated control method for a variable refrigerant flow (VRF) system and a humidifier system. Rather than simply setting a temperature control, a DQN was used to derive the integrated control method for VRF and humidification.

Current DRL approaches primarily treat the non-stationarity of building environments in a passive fashion, hindering their practical applications. Potentially, DRL can relearn the system dynamics, although this is time-consuming. To overcome this issue, Deng et al. [113] proposed a new HVAC control method (termed unsteady DQN), which combines active building environment change detection with DQN. This

method aims to actively identify the change points of the building environment to eliminate the non-stationary. It was found to be superior to DQN in both single- and multi-zone control. Instead Lork et al. [70], used Bayesian convolutional neural networks (BCNNs) to model the performance and uncertainty of AC from aggregated data, and combined with Q-learning for set point decision making. MPC approaches non-stationarity via the nonlinear dynamics, and robust as well as stochastic MPC were developed precisely to take into account uncertainties in the dynamics. The authors in Refs. [70,113] fail, however, to make comparisons to these approaches, so it is difficult to fully assess the merits of their approaches.

Zhao et al. [49] proposed a hybrid model-based DRL method to solve the HVAC control problem. This method uses both a knowledge driven model and a data driven model in the whole learning process. Ultimately, the protection mechanism and adjustment incentive method are used to further reduce the learning cost and overcome the shortcomings of low learning rate and high learning cost of DRL. Gao et al. [52] employed a deep MLP and Bayesian regularization to predict the thermal comfort of residents, which was combined with DDPG for thermal comfort control. A comprehensive comparison was made between DDPG and the widely used DQN method by Du et al. [76], and their results suggested that DDPG is more effective at dealing with continuous motion space and can achieve higher economic benefits while maintaining user comfort.

The advantages of the combination of MPC and ML mentioned above have broad prospects for optimal control of HVAC. Numerous efforts in this area have achieved good results. The performance of MPC depends crucially on the accuracy of the prediction model. Nagpal et al. [51] used a robust MPC controller for the indoor climate of buildings based on a linear state-space model of the indoor environment with polytopic uncertainties, leading to a robust control problem in the form of LMIs, forming a semidefinite programming problem. The performance was 24% better than that of a nominal MPC approach, although no other comparisons were made. This approach is potentially very powerful, in that it leverages the advantages of robust MPC with a principled state-space representation of the environment. It is worth exploring the approach further for an optimal balance of computational cost and accuracy, given the plethora of state-space models and optimization approaches available.

Lee et al. [55] developed a self-tuning HVAC controller, with a personalized thermal preference model, the transfer of thermal conditions and model predictive control (MPC) to form a closed loop. Based on occupant feedback and the personalized thermal preference model, Bayesian clustering and an online classification algorithm are used to determine the local operating temperature boundary of each radiant floor circuit. Ellis and Chinde [94] point out that the relationship between temperature setpoints and HVAC load is nonlinear, while the thermal dynamics of a building are linear or weakly nonlinear. This is used as a motivation for a purely data driven MPC strategy and in particular they adopt an economic MPC (EMPC) approach with an LSTM encoder-decoder to predict the temperature and cooling rate for a HVAC system. EMPC directly embeds an economic cost function into the MPC framework, which typically involves a quadratic cost function that is insensitive to economic costs. The model was validated using a case study generated in EnergyPlus. Starting from the definition of inputs and outputs, the model is trained and automatically embedded into the EMPC framework for real-time estimation and control. This novel strategy only partially offsets the total cost of build, configuration, and training of EMPC models, a major reason why they have not been adopted in practice.

Fu and Zhang [98] proposed a hybrid control method based on TD3 and MPC. By pre adjusting the building temperature during off peak hours to reduce the function approximation error, they effectively avoid overestimation and improve the accuracy of energy consumption prediction and control of a HVAC system. A model-based reinforcement learning method was proposed by Zhang et al. [103], which uses ANNs to learn the system dynamics. MPC is then adopted to control the system by using the learned system dynamics and random sampling method.

Compared with model-less RL, this method improves the training efficiency. Model-based RL has not been investigated as much as model-free RL, for reasons already stated. Nevertheless, there are reasons to believe that a model-based RL, although perhaps not as transferrable as model-free RL, can be superior for a given system with a well-constructed model. This is certainly worthy of exploration given the general reluctance of industry to adopt RL approaches.

It is worth mentioning that non-machine-learning approaches continue to be developed. In Ref. [101], Xie et al. develop a distributed model predictive control algorithm based on the alternating direction method of multipliers (ADMM) to deal with system uncertainty and space-time coupling constraints. ADMM essentially deconstructs the optimization problem into sub-problems that are easier to solve. The algorithm has high scalability with the increase of the number of regions, and can protect the privacy of users. Li and Zhang [102] also took a non-machine learning approach based on a hybrid optimization algorithm composed of improved particle swarm optimization (IPSO) and Newton Raphson (NR) for a large-time delay single district heating system. The method was used for online optimization within an MPC framework applied to single district buildings. A distributed MPC method was proposed for multi area buildings with hydraulic coupling. It relies, however, on a complex model, unlike the model of Xie et al. [101].

3.4. Objectives for machine learning in HVAC

According to Table 2, 95.59% (65 papers) of the literature considered energy savings for HVAC, 89.71% (61 papers) of the literature considered optimization of thermal comfort, and 85.29% (58 papers) optimized the energy savings and thermal comfort together. It is evident that researchers have attached great importance to these two criteria.

The indoor environment affects residents' health, productivity [140, 142] and happiness [143,144]. In the architectural literature, there are several recognized comfort factors [145]. These factors include thermal comfort, indoor air quality, lighting and noise. Thermal comfort is widely considered to be the most important aspect among these different indoor environmental factors [146,147]. Ensuring thermal comfort and working performance is one of the main objectives of building environment regulation systems. Air quality, lighting and noise are less studied than thermal comfort. Fig. 8 shows the number of articles addressing different comfort factors, amongst those included in this review. It can be seen that most of the papers only optimize the thermal comfort, while a much smaller number consider the air quality. None of the articles address sound and visual factors. This immediately opens up a direction for future research, with HVAC systems optimized with a comprehensive consideration of all comfort factors.

Fig. 9 shows the factors that affect building occupants' thermal comfort. There are six main factors that affect human thermal comfort. These six basic factors can be divided into two categories. The first category is environmental factors, including air temperature, relative humidity, air velocity and mean radiation temperature. The second

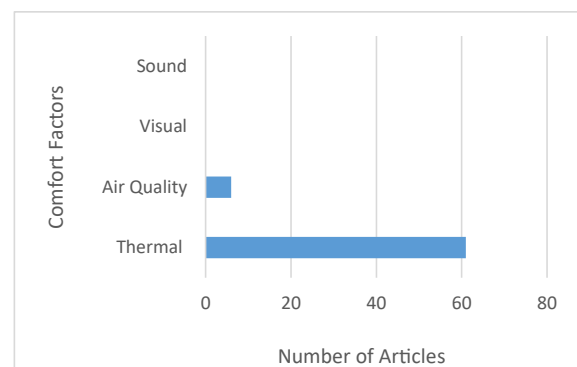


Fig. 8. Numbers of articles addressing different comfort factors.

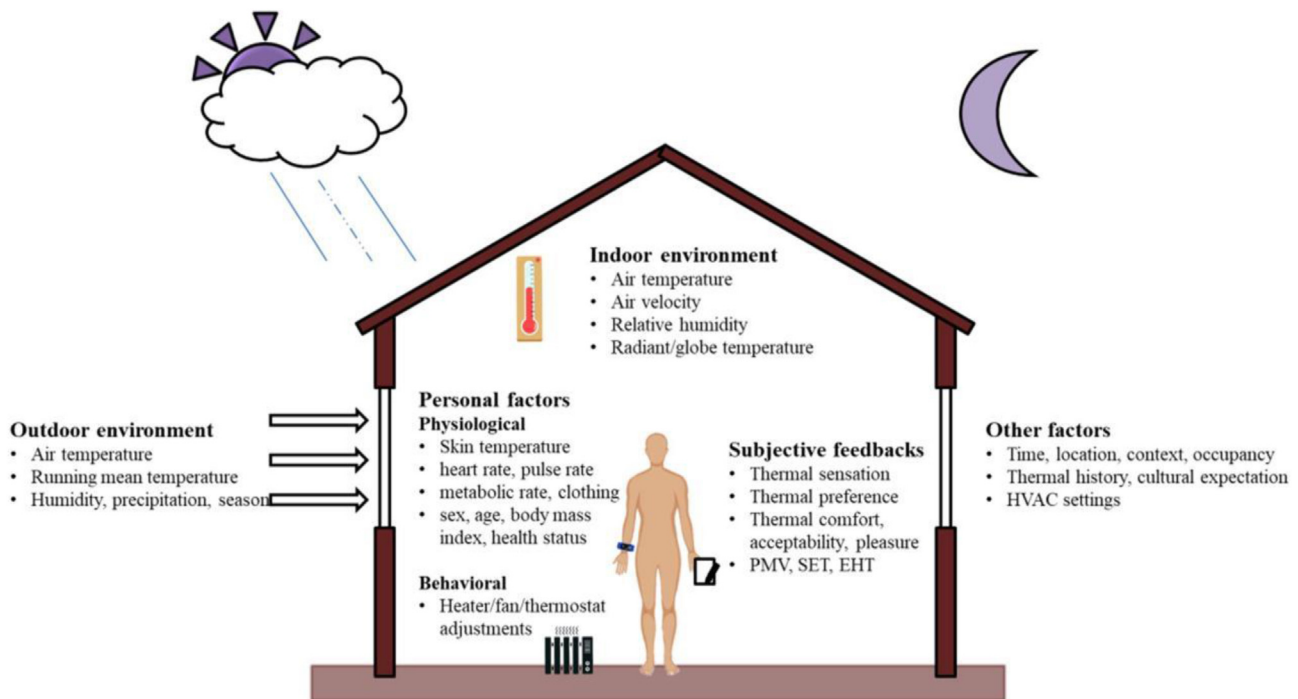


Fig. 9. Factors affecting building occupants' thermal comfort [148]. Reprinted with permission from Elsevier.

category is human factors, including clothing thermal resistance and metabolic rates. These factors need to be transformed into thermal comfort evaluation metrics when machine learning is applied to HVAC system optimization.

A building thermal comfort evaluation model is the main way in which to judge a thermal environment. Thermal comfort evaluation models can be divided into two categories: thermal safety models and thermal balance models. A thermal safety model is used to ensure the safety of workers during operation, so this paper will not discuss such models. It should be noted that predicting thermal comfort is challenging since it is affected by several factors, including the environment and an individual's physiology and even psychology. Metrics such as metabolic rate or skin temperature may be used to quantify some of the factors but such measurements are not always readily available or practical and cost effective to gather for each occupant.

In [50], Gao et al. propose a framework based on DRL, which defined thermal comfort control as a cost minimization problem, and designed an ANN based thermal comfort prediction model for residents. The simulation results showed that the thermal comfort of the members was improved by 13.6%. The thermal comfort cost (TCC) is introduced by Lymporopoulos and Ioannou [89], and the acceptable range of the optimal set point of the area temperature is determined based on the specified value of the predicted mean vote (PMV). In Ref. [97], Gao et al. use a DNN method with Bayesian regularization to predict the thermal comfort of residents by considering different influencing factors.

Zhu et al. [85] develop a rapid prediction model for indoor environmental parameters (i.e. air pollutant concentration, temperature, and humidity), and ultimately achieve online control of a HVAC system by sequentially adjusting the optimal hourly air exchange rate, supply air temperature, and humidity. In the study of hourly air exchange rate, CFD simulations were conducted to determine indoor CO₂ concentrations, temperature and humidity based on different input variables, ultimately to establish an initial CFD database. The initial CFD database was expanded using low dimensional linear models (LLMs) for temperature and ventilation. On the basis of the LLMs, an ANN was used to learn the map from inputs to CO₂ concentration, temperature and humidity.

Zhou et al. [149], developed a personalized thermal comfort model in which nonlinear regression is used to predict personalized metabolic

rates and skin temperatures, which are fed into a physical heat balance model based on feedback from occupants. The solution algorithm enables the personalized thermal comfort model to be dynamically updated through online calculation. With the continuous updating of the model, changes in the residents' comfort can be tracked, and a high prediction accuracy can be maintained. The authors also considered other regression techniques, including K-nearest neighbors (KNN), support vector regression (SVR), and decision trees (DT), finding that KNN gave the best performance in terms of predicting the thermal vote.

In [150], Aparicio-Ruiz et al. proposed a general framework based on a dynamic adaptive comfort algorithm. The authors use KNN to learn the thermal comfort temperature of residents relative to the average operating temperature, so as to determine the appropriate indoor temperature range. Ghahramani et al. [151] developed a model for human face heat comfort based on a Hidden Markov Model (HMM). In this study, infrared thermal imaging technology is used to provide real-time information about the thermal comfort of residents. The HMM based learning algorithm has three hidden states (uncomfortable warmth, comfort and uncomfortable coolness), and uses discrete modules to form the observation state of continuous infrared measurements. The effectiveness of the method was verified by experiments.

Classification methods can also be used to develop thermal comfort models. In Ref. [74], an image classification model was trained based on a CNN to classify the clothing grade of residents, and a LWIR camera with face detection was used to obtain the average facial skin temperature of residents for the classification input. In Ref. [152], Salleh et al. compared six classification methods and PMV for predicting thermal comfort based on six thermal factors (air temperature, average radiation temperature, relative humidity, wind speed, clothing insulation and metabolic rate). They found that KNN worked best (80.41% accuracy), followed by DT, a CNN, SVM and kernel Naïve Bayes. In addition, any data driven method was superior to the results obtained by modeling thermal comfort with the PMV model (22.68% accuracy). In this way, the results in Ref. [152] show that the ML not only has the advantage of being able to build models with only historical data, but it can also provide accurate thermal comfort models, and can even exceed the widely accepted PMV in terms of accuracy.

Among the factors related to residents, building occupancy rate is the most critical factor and also the basis for studying residents' behavior.

Fig. 10 illustrates the definition and impact of building occupancy rate. The occupancy rate of a building includes its existence status, number of occupants, and trajectory, which indicate whether the occupants are in a certain space of the building, the number of occupants, and how they move. Irregular and localized living have a significant impact on indoor environment and energy consumption. HVAC control requires a certain amount of pretreatment time, so the prediction of building occupancy is a very important factor for HVAC system optimization. Too low or too high an indoor temperature will cause thermal discomfort when occupants arrive, while also potentially wasting energy. Therefore, good occupancy prediction is crucial to the optimization of the system. However, the prediction of occupancy rate is complex, and the occupancy rate in buildings is often highly dynamic. With time, the flow of occupants can be highly random. In recent years, the concept of randomness [153–155] and diversity of building occupancy has been widely studied and recognized [156–158].

According to Table 2, 18 articles (26.47%) considered occupancy rate and user activity, and most of the studies did not consider the impact of occupancy rate on HVAC optimization, which suggests that a great deal more can be done on this vital topic. At present, there are several challenges in the research of occupancy rate, including data collection and data set establishment. Although a variety of data collection methods have been applied, such as occupancy and vacancy time questionnaires [160], motion sensors [161–163], radio frequency identification (RFID) sensors [164], door sensors [165], Bluetooth beacons [166] and mobile phone records [167], it is still challenging to obtain occupancy related data in buildings due to accuracy and privacy issues. However, with the development of the Internet of Things, data acquisition can be improved. A new challenge is how to mine and process these data. Another challenge is to consider the natural characteristics of building use. The prediction accuracy can be improved by combining natural features such as time and space features with machine learning methods. In addition, different event related drivers must be considered.

Predicting occupancy can be approached in different ways, including classification methods to predict occupancy states or regression to predict departure/arrival times. In Ref. [168], Esrafilian-Najafabadi and Haghighat analyzed the impact of different approaches on the performance of residential based HVAC control systems. The effectiveness of using different ML techniques in predictive control systems, namely DT, KNN, a MLP and a Gated Recurrent Unit (GRU), was also evaluated. The authors found that the classification methods were generally the best

performing, with either KNN or DT being the best, depending on whether the objective was thermal comfort or energy savings. There are, however, many specialized time-series methods that are arguably more appropriate for occupancy prediction, being as it is a sequence problem. Among the many methods are linear state-space models such as autoregressive integrated moving average (ARIMA) models and their variants, as well as Kalman filters. There are also nonlinear state-space models such as HMMs and Gaussian process dynamical models, as well as agent-based models. Moreover, purely supervised ML approaches with data embeddings can also be used. These specialized methods have not been explored to any great extent in HVAC research, with most of the focus placed on RNNs, which are more suitable for natural language processing rather than time-series problems.

As alluded to in the previous section, in RL, the occupancy can be modelled as MDP, namely we seek to maximize the return in a MDP. An MDP is defined by the set of states of the system, the set of actions that can be executed, a probability distribution over the states from a given state and with a given action, and finally a reward function that assigns a reward to transitions between states for a given action. Deng and Chen [57] used a MDP to simulate the behavior of residents, combined with Q-learning to control occupancy comfort. The perfect occupancy prediction assumption is considered in Ref. [77] to eliminate the impact of uncertainty related to occupancy prediction and estimate the upper limit of system performance. A rule-based control is implemented alongside a deep MLP that uses historical weather conditions to learn a map between these conditions and the lag time (or preconditioning time) of a HVAC system. The results show that occupancy-based control systems can increase energy consumption by up to 10%, while using a conservative setback can compromise financial savings, leading to long payback periods with the use of smart thermostats. These negative impacts highlight the need to consider peak demand management when developing occupancy-based control systems.

Wang et al. [112] used a vision-based camera for identifying the activities of indoor residents and equipment, and for calculating the real-time value of internal heat from residents and equipment. Feeding this information into energy and thermal models, thermal dissatisfaction was predicted using ANNs. Vosughi et al. [90] used a Markov jump linear system model to capture the thermal process of buildings, HVAC systems, and occupant movement, which they combined with a classical PID controller. This method can respond to the location changes of building occupants in real time. The proposed control scheme used real-time

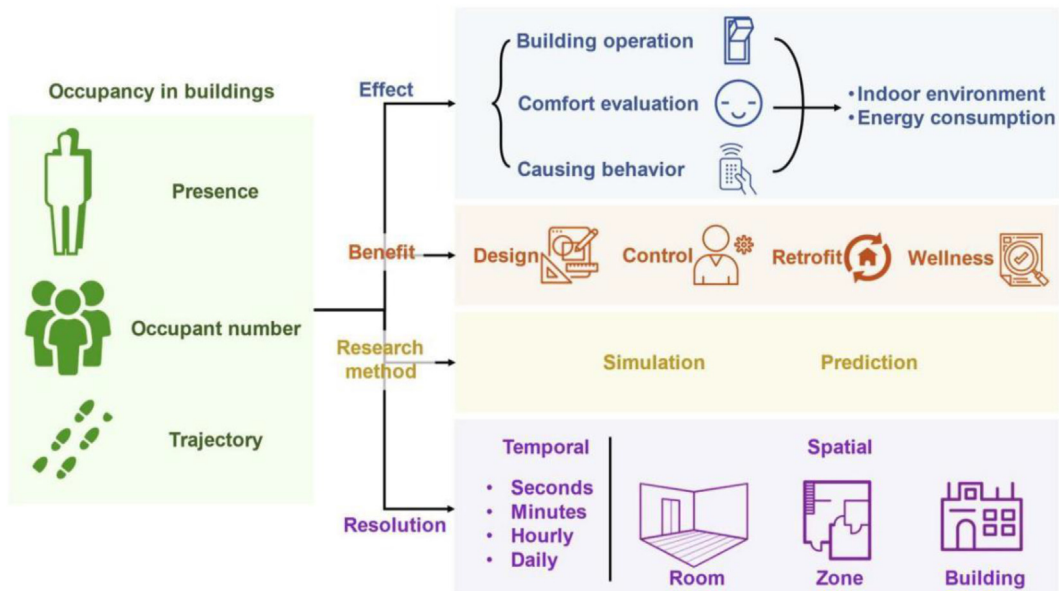


Fig. 10. Definition and concept of building occupancy [159]. Reprinted with permission from Elsevier.

temperature and occupant position sensing, as well as a combined model of building thermal process and occupant movement. A Q-learning control strategy was proposed by Chen et al. [80] for a HVAC system and window operation involving natural ventilation. The authors compared the results to a rule-based control and demonstrated significant improvements. Window activity is an important human factor, but at present it is usually neglected.

According to Table 2, 27 papers have provided specific energy saving percentages, among which the research results of Zhu et al. [85] suggest that the total energy consumption of HVAC systems can be reduced by up to an impressive 35%. Although the comparison strategy and simulation strategy used in each study are different, ML can achieve considerable energy savings for HVAC systems. Considering electricity price can further reduce the cost for users. The price of electricity is high in the peak use period, when HVAC is most frequently used. The optimization strategy proposed by.

Kim et al. [78] obtains the optimal solution under various conditions of a given electricity price and building heat conditions. The authors use a deep nonlinear auto-regressive network with delayed inputs to directly determine the optimal DR scheduling. The proposed method maintains the load transfer ability of the air conditioning system to time-varying electricity price. Yoon et al. [79], used an ANN model to establish an optimization problem for price-based DR, and the problem was solved repeatedly to generate optimal HVAC load plan data for different electricity prices and building environments. By using the ANN metamodel (or surrogate model) to replace the DR optimization problem, the optimal retail price can be achieved by using a single-layer decision-making structure. This study simplifies the pricing optimization problem, and therefore makes the HVAC system easier to implement in large distribution networks.

RL can be time consuming by virtue of the heuristic search. Moreover, especially in the early learning stage, the exploration-vs-exploitation mechanism can be unstable, and large step sizes can lead to poor control [106]. An alternative paradigm is to develop data-driven HVAC and building models offline first and to train the RL agent with data obtained from these models, as was done in Refs. [52,108]. Similarly, Jang et al. [68] developed a model for indoor temperature control in a commercial building by establishing LSTM models for three sub-systems (thermostat controller, HVAC unit, and building envelope), given next-day forecasts of the electricity price. The ANN models were integrated to establish an optimization problem to schedule the temperature set-point. To avoid overfitting, the ANNs were trained online. Compared to predetermined set-point scheduling, a physics-based model and other ML based strategies, their model was shown to be effective in terms of cost.

The hybrid TD3-MPC control method of Fu and Zhang [98] considers both the electricity price and thermal comfort. The function approximation error is reduced and the costs lowered by pre-adjusting the building temperature during off-peak hours. A modified time of use (TOU) tariff was used in Ref. [93] to simulate the total operating cost (TOC). With developments in building intelligence, renewables and energy storage, photovoltaic, wind, battery, thermal storage and other technologies are increasingly used to supply power to buildings. The optimization of HVAC systems in different types of power grids and in different settings is one of the directions that needs to be explored in the future.

3.5. Fault detection and diagnostics

California conducted air conditioning inspections on 13000 air conditioning systems, and the results showed that 65% of residential units and 71% of commercial units needed repairs [169]. Insufficient maintenance, degraded component functionality, installation errors, and control failures seriously affect the performance of HVAC systems [170]. Due to the complexity and lifespan of HVAC systems, they often encounter various malfunctions and problems, such as refrigerant leaks, valve failures, and circuit problems. Traditional HVAC fault detection

methods often rely on the experience and intuition of professional technicians, which has many limitations, such as low detection efficiency, unstable detection results, and high costs. Research has shown that due to building system failures and improper control, 15%–30% of the energy may be wasted [171]. Therefore, fault detection and diagnostics (FDD) or automated fault detection and diagnostics (AFDD) could be crucial for ensuring reliable system operation and avoiding energy waste. As a consequence, researchers and engineers are increasingly exploring the use of machine learning algorithms to solve the problem of HVAC fault detection. Here, “fault” implies deviations from normal operation in at least one measurable quantity, “detection” is the task of discovering the fault, and “diagnosis” refers to the identification of the origin of the fault as well as its location.

Model-based FDD approaches appeal to the underlying physics insofar as it can be understood, while rule-based approaches extract a list of rules for decision making on the basis of expert knowledge and known underlying physics, which will change from system to system. Machine learning approaches are more efficient and scalable; machine learning algorithms can quickly process a large volume of data, discover hidden patterns and accelerate the process of fault detection and troubleshooting. Machine learning algorithms for FDD can be adaptive, with adjustments made based on real-time data to cover different environments and situations. Furthermore, machine learning algorithms can allow for visualization, in order for engineers to analyze and make decisions regarding faults. Finally, no complex model of the system is required, making the approaches much more adaptable and eliminating the need for an accurate, validated model. This also avoids the problem of parameter value drift with time, and the computational expense of running the model. Table 3 summarizes a selection of the main machine learning studies related to FDD of HVAC systems.

One of the methods by which faults can be detected is statistical process monitoring (SPM), which draws on multivariate data from the process to find anomalies, usually based on principal component analysis (PCA) or partial least squares. PCA is a commonly used dimensionality reduction technique for data that lie in a high-dimensional ambient space. PCA finds an approximate representation of the data in low-

Table 3

Summary of machine learning algorithms applied to fault diagnosis and detection.

HVAC Components	Machine Learning Algorithms	Reference
Chiller	PCA	[172–174]
	SVM	[175–178]
	Linear Discriminant Analysis (LDA)	[179]
	Linear Classifier	[180,181]
	Bayesian Network Classifier (BNC)	[182,183]
	Kernel PCA	[184]
	Deep Belief Network	[185]
	CNN	[186]
	Generative adversarial network	[187]
	PCA	[188]
Air-Handling Unit (AHU)	SVM, KNN	[189]
	TL	[190]
	ANN	[191]
	PCA, RF, Time series anomaly detection	[192]
	SVM, ANN, RF	[193]
	SVM, RNN	[194]
	ANN	[195]
Variable Air Volume (VAV) unit	PCA	[196]
Variable Refrigerant Flow (VRF) system	BNC	[197]
	Decision Trees (DT)	[198]
Terminal unit	SVM	[199]
Whole system	ANN	[200]
	Image-based TL	[201]
	SVM, TL	[202]
	DL	[203]

dimensional subspace of the ambient space. The data (typically centered by the sample mean) is orthogonally projected onto a set of principal directions, which are found by an eigenanalysis of the sample covariance matrix. The principal directions represent directions of decreasing variance in the data. The components of the data in this new basis are referred to as principal components. The data can be reduced in size by restricting to a subspace corresponding to those principal directions that capture a specified fraction of the total variance. In HVAC FDD, normal operating data is used to find the principal directions. New measurements can be projected onto a subspace defined by a specified number of the directions and the residual (difference between the projection and the actual data point) can be subjected to different statistics, such as a Q or Hotelling statistic to determine statistical significance.

Movahed et al. [192] used PCA to reduce the dimensionality of the data, accounting for 95% of the data variance. Subsequently, methods of time series anomaly detection and RF were used to reduce false positives and to account for data imbalances; the authors rightly emphasize that insufficient attention has been paid to false positives in HVAC FDD research. The method proposed by Simmini et al. [184] uses kernel principal component analysis (KPCA) in place of PCA. The motivation for nonlinear dimension reduction (manifold learning) techniques such as KPCA, Isomap, diffusion maps and local tangent space alignment is that for some data sets a linear subspace approximation (e.g., PCA or classical multidimensional scaling) is not possible to find. In such cases, nonlinear dimension reduction techniques essentially map the data to a feature space, while preserving a metric (distance) in the original space. The methods differ in how this metric is defined in the original space, while in the feature space a Euclidean metric is always used. For example, Isomap uses a geodesic distance whereas diffusion maps use a transition probability defined by identifying data points with nodes on a graph and constructing a Markov chain. An alternative, and more natural interpretation of KPCA is that linear PCA is performed in the feature space, with no explicit specification of the feature map. Instead, the eigenvalue problem in feature space can be written entirely in terms of a kernel (which is specified) and kernel substitution is employed. In HVAC fault detection and diagnosis, KPCA and related methods can help to better understand the operating status of the system and diagnose potential faults when there are highly non-linear relationships in the data that cannot be rationalized using PCA. To the best of our knowledge, only KPCA has been used in the context of HVAC systems, even though there are many other methods available, and it is difficult to know a-priori which method is best suited for a given data set. There are many questions that need to be answered in future research, including the conditions under which linear methods fail, the efficacy of nonlinear methods in such cases and comparisons between the different methods for different data types.

A key aspect of machine learning algorithms is the acquisition and processing of data. Data quality is crucial for the accuracy and credibility of the final results, and labeled data is even more difficult to obtain. This is because collecting labeled data for fault conditions is time-consuming, expensive, and sometimes even dangerous. In addition, HVAC systems are often complex and dynamic, and different systems may have different characteristics and behaviors. This makes it difficult to develop a universal FDD model suitable for different HVAC systems. With the development of technology, there are increasingly more channels for data acquisition.

Hosamo et al. [193] proposed a digital twin predictive maintenance framework for AHU to overcome the limitations of facility maintenance management (FMM) systems currently used in buildings. This framework is designed to predict component status, maintain or repair components in advance, and improve the service life of AHU components. The digital twin technology includes data integration and a data flow process among building information modeling, IoT sensor networks and facility manager systems. The proposed framework includes three main components, namely data acquisition, the predictive maintenance step and a BIM model, which is used for visualization and monitoring. Machine learning

in the form of ANNs, SVMs, and DTs was used to classify faults, after selecting inputs using a sensitivity analysis. Bezyan et al. [194] used normal operating data to build models for two key temperature sensors relating to an AHU, both models based on SVR and using various temperature and flow rate values as inputs. These models were used to predict the expected values of temperature, which were compared to measured values to detect anomalies on the basis of the residuals or differences. Separate RNN models were used to predict whether each of the inputs was anomalous, rather than, or in addition to the temperature output. Once all anomalies were found, a rule-based method was used to detect a fault.

Although data is available via multiple channels and can be used for FDD as in Ref. [193], high-quality data sets are usually expensive and difficult to obtain, and it is even more difficult to label the obtained data, which requires in-depth understanding of HVAC system behavior and thermodynamic principles. Moreover, there are issues associated with unbalanced data sets, meaning the target classes have an uneven number of examples. Albayati et al. [189] developed a semi-supervised learning approach by first using clustering to group faults, followed by locating the k unlabeled nearest neighbors to a given set of labelled training points (Euclidean distance based). They then labelled these points according to which training point was closest, while restricting the search for neighbors such that they fell within the same cluster. In this way, the size of the training set for fault classification was increased and the problem of imbalanced data addressed, leading to greater accuracy. They applied the method to a HVAC system in an industrial facility in Connecticut, using SVM for classification, with up to 95.7% accuracy.

Another way to improve results by augmenting a data set is to use transfer learning, leveraging knowledge from other HVAC systems. Fan et al. [191] proposed a new image-based transfer learning framework to solve the challenges of multi-source data compatibility in the field of architecture, and discussed the value of transfer learning in cross domain FDD tasks. They effectively integrated table data from different building systems, proposing a two-dimensional image conversion method that converts table data into two-dimensional images, followed by the extraction of features for knowledge transfer using CNNs. Various transfer learning strategies were then adopted to migrate knowledge from the source to target task (system). The accuracy was found to increase from between 2.4% and 22.2%, depending on the volume of training data available; namely, the lower the volume of data the higher the accuracy gain, for obvious reasons. This suggests that transfer learning is only valuable (given its associated time costs) when limited data is available for the target task.

Viol et al. [190] pointed out that most of the research on transfer learning for HVAC systems has focused on learning algorithms, ignoring the importance of appropriate domain similarity analyses of the available data. Their main contribution is the application of a pre-training filter that discards dissimilar samples. Seven feature-based and instance-based transfer learning approaches were tested, and the results on AHU system data revealed that negative transfer is a serious issue (the results deteriorate with transfer learning). In some cases the filter able to offset this issue, in one particular case raising the F score from 0.29 to 0.9. These results are preliminary, however, and more research into these phenomena, including the detailed inter-relationships between the transfer learning method, the volume of data, and the dissimilarity of the data sets, is required.

Aside from the aforementioned methods, a wide variety of other machine learning algorithms have been applied to FDD of HVAC systems, such as deep belief networks [185], Bayesian network classifiers [182, 183], and generative adversarial networks [187]. The application of machine learning in this area can help to better understand the operating status of systems and diagnose potential faults, improve system efficiency and reliability, and reduce costs and risks. When applying machine learning, special attention needs to be paid to the quality and quantity of data, as well as the selection and tuning of models. In addition, the various methods can be used within a semi-supervised framework and

alongside transfer learning to further improve their accuracy and reliability by incorporating more data, and allowing for imbalanced and missing data, which is the usual scenario in real-world applications.

4. Future research directions and conclusions

HVAC systems account for more than 40% of building energy consumption, and are one of the main sources of carbon dioxide emissions. Machine learning methods could have a transformative impact on the optimization and control of HVAC systems, as well as on building design and fault diagnosis and detection. Despite over a decade of research into this area, however, it is fair to say that there are few practical examples of their adoption. The selection of appropriate machine-learning methods depends on several factors, such as the application, the data type, data quantity and quality, calculation cost and calculation complexity. At present, the machine learning methods used in HVAC system optimization are mainly supervised learning and reinforcement learning. Many studies, however, adopt a mixed approach, combining these methods with unsupervised methods (clustering, dimension reduction), and/or within semi-supervised and transfer learning approaches.

The sheer complexity of HVAC optimal control problems, involving dynamic thermal and occupancy models, large and complex data sets, electricity pricing, and a complex multi-objective constrained optimization would suggest a complex solution, drawing upon various different techniques to form submodels with various options to facilitate transferability. It could also be that such a transferability is not possible, at least not when the applications are vastly different in type and scale.

DNN approaches have become widely used, and perhaps overused in recent years. DNNs are extremely powerful and versatile, capable of high accuracy, applicable to supervised (classification and regression) tasks, unsupervised tasks such as feature extraction and time series or sequence problems using RNNs. Convolutional or dense layers can be used to extract features for improved classification, especially when the data set is not linearly separable. On the other hand, they have a number of shortcomings, such as the extremely large number of parameters and associated danger of overfitting, the difficulty in obtaining stable solutions (high model variance), the difficulty in finding optimal architectures given the vastness of the search space, and the times taken to design and validate DNNs. Much more comparisons are required to other equivalent approaches such as GP models, SVM/SVR, and kernel regression to determine under which conditions they perform better.

Another vital aspect for which DNNs, notably RNNs, have been used is the dynamic modelling of buildings, e.g., occupancy and load prediction. While RNNs are unrivalled in natural language processing, namely predicting words in a sequence, they are generally not favored in time-series analysis. The main issues are their complexity, lack of explainability, and absence of confidence intervals. Alternatives such as hidden Markov models have already been explored for HVAC systems. However, there are many other linear and nonlinear state-space methods that could provide better predictions than RNNs, and these have yet to be considered or compared in the HVAC literature.

DRL is widely used in HVAC optimal control, combining the benefits of RL with the power of deep learning. However, it is fair to say that the various competing methods, especially DRL, MPC and robust MPC, rule-based methods and mixed methods such as nonlinear dynamics and MPC have not been systematically compared in the literature. The vast majority of studies provide very few comparisons. At the present time, therefore, it is not possible to say which of these approaches would work best for a particular type of system. It is doubtful that one approach works best for all conditions, and the performance of each for different types and volumes of data, different systems, and different objectives requires exploration.

The two most important factors in the optimization of HVAC systems are energy savings and user comfort. In terms of energy conservation, most studies do not consider electricity prices or distributed and smart grids. The price of electricity fluctuates, with the peak period of HVAC

use often coinciding with the period of peak electricity price. Considering electricity price in the optimization process can further reduce costs, and should therefore feature more prominently.

Comfort includes four aspects: thermal comfort, air quality, noise and visual factors. At present, HVAC optimization mainly focuses on thermal comfort, and some studies consider air quality, such as carbon dioxide concentration, while very few have considered noise and visual factors. Air quality and noise should be incorporated more often, especially in high-occupancy and specialized settings such as educational buildings, offices, libraries and museums. In terms of user thermal comfort, thermal insulation and metabolic rate are parameters that are difficult to accurately measure at present, and need to be further studied. In addition, most studies do not consider the occupancy rate and residents' activities. Models for occupancy rate and activity are a major avenue for further investigation given the importance of these factors in devising accurate and reliable control policies.

In the literature reviewed in this paper, many of the methods were verified by simulation, lacking real-world experiments. Simulated environments are simplifications of the real world, and while they may act as a reference they should not be used alone for the purposes of validation. A final challenge relates to the development of the Internet of Things, and the attendant improvements in data acquisition. Using this raw data is not feasible, not only due to its size but also the redundancy of information that it will contain. How to preprocess and mine this data (filter and de-noise, extract features, reduce the size, etc.) is a major challenge, for which machine learning and more broadly data science is ideally suited.

Funding

This work was supported by the Creative Research Groups of the National Natural Science Foundation of China (No. 52021004).

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Nomenclature

ADMM	Alternating Direction Method of Multipliers
AHU	Air Handling Unit
ANN	Artificial Neural Network
BEM	Building Energy Management
BMS	Business Management System
CFD	Computational Fluid Dynamics
CNN	Convolutional Neural Network
CT	Classification Tree
DDPG	Depth Deterministic Policy Gradient
DNN	Deep Neural Network
DQN	Deep Q-network
DR	Demand Response
DRL	Deep Reinforcement Learning
DT	Decision Tree
EWB	Electric Water Heating
FDD	Fault Detection and Diagnosis
FMM	Facility Maintenance Management
GA	Genetic Algorithm
GPR	Gaussian Process Regression
GRU	Gated Recurrent Unit
HVAC	Heat Ventilation and Air Conditioning
HMM	Hidden Markov Model
KNN	k-Nearest Neighbor
KPCA	Kernel Principal Component Analysis
LB-TSA	Learning Base Token Scheduling Algorithm

LMI	Linear Matrix Inequality
LSTM	Long Short-Term Memory
LR	Linear Regression
MAPE	Maximum Absolute Percentage Error
MILP	Mixed Integer linear Programming
MJLS	Markovian Jump Linear System
MPC	Model Predictive Control
MLP	Multilayer Perceptron
ML	Machine Learning
MTL	Multi-task learning
MUB	Mixed-Use Building
PCA	Principal Component Analysis
PID	Proportional Integral Derivative
PMV	Predicted Mean Vote
PPD	Percentage of People Dissatisfied
PPO	Proximal Policy Optimization
PSO	Particle Swarm Optimization
RDT	Regression Decision Tree
RF	Random Forest
RL	Reinforcement Learning
RNN	Recurrent Neural Network
SVM	Support Vector Machine
SVR	Support Vector Regression
TCC	Thermal Comfort Cost
TD3	Twin Delayed Deep Deterministic Policy Gradient Algorithm
TRPO	Trust Region Policy Optimization
TL	Transfer Learning
VRF	Variable Refrigerant Flow
WRLS	Weighted Recursive Least Square

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